Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

2079

Bachelor of Science in Computer Science and Information Technology

Course Title: **Design and Analysis of Algorithms**

Code No: **CSC325** Semester: **5th Semester**

Full Marks: **60** Pass Marks: **24** Time: **selection algorithm and analyze its complexity.2.Write the dynamic programming algorithm for matrix chain multiplication. Find the optimal parenthesization for the matrix chain product ABCD with size of each is given as A■■■■, B■■■■■, C■■■■■, D■■■■■.3.What do you mean by Backtracking? Explain the backtracking algorithm for solving 0-1 knapsack problem and find the solution for the problem given below:Items →1234Weight (w■) →6235Profit (P■) →45107and capacity of knapsack: 10 kgSection BAttempt any Eight questions[8×5=40]4.Explain the iterative algorithm to find the GCD of given two numbers and analyze its complexity.5.Trace the quick sort algorithm for array A[ ] = {15, 7, 6, 23, 18, 34, 25} and write it's best and worst complexity.6.Generate the prefix code for "CYBER CRIME" using Huffman algorithm and find the total number of bits required.7.Solve the following recurrence relations using masters methoda) T(n) = 2T(n/2) + kn², n > 1= 1, n = 1.b) T(n) = 5T(n/4) + kn, n > 1= 1, n = 1.8.Find the edit distance between the string "ARTIFICIAL" and "NATURAL" using dynamic programming.9.Find the MST from following graph using Kruskal's algorithm.10.Solve the following linear congruences using Chinese Remainder Theorem.x ≡ 1 (MOD 2)x ≡ 3 (MOD 5)x ≡ 6 (MOD 7)11.Define tractable and intractable problem. Illustrate vertex cover problem with an example.12.Write short notes on:a) Best, Worst and average case complexityb) Greedy Strategy**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Explain the divide and conquer strategy for problem solving. Describe the worst-case linear time selection algorithm and analyze its complexity.

2. Write the dynamic programming algorithm for matrix chain multiplication. Find the optimal parenthesization for the matrix chain product ABCD with size of each is given as A■■■■, B■■■■■, C■■■■■, D■■■■■.

3. What do you mean by Backtracking? Explain the backtracking algorithm for solving 0-1 knapsack problem and find the solution for the problem given below:Items →1234Weight (w■) →6235Profit (P■) →45107and capacity of knapsack: 10 kg

Section B

Attempt any Eight questions[8×5=40]

4. Explain the iterative algorithm to find the GCD of given two numbers and analyze its complexity.

5. Trace the quick sort algorithm for array A[ ] = {15, 7, 6, 23, 18, 34, 25} and write it's best and worst complexity.

6. Generate the prefix code for "CYBER CRIME" using Huffman algorithm and find the total number of bits required.

7. Solve the following recurrence relations using masters methoda) T(n) = 2T(n/2) + kn^2, n > 1= 1, n = 1.b) T(n) = 5T(n/4) + kn, n > 1= 1, n = 1.

8. Find the edit distance between the string "ARTIFICIAL" and "NATURAL" using dynamic programming.

9. Find the MST from following graph using Kruskal's algorithm.

10. Solve the following linear congruences using Chinese Remainder Theorem.$x \equiv 1$ (MOD 2)$x \equiv 3$ (MOD 5)$x \equiv 6$ (MOD 7)

11. Define tractable and intractable problem. Illustrate vertex cover problem with an example.

12. Write short notes on:a) Best, Worst and average case complexityb) Greedy Strategy